

Multiple Choice (10 marks)

1. The message must be encrypted at the sender site and decrypted at the
 - (a) Sender Site
 - (b) Site
 - (c) Receiver site
 - (d) Conferencing
2. Let A be a finite set with m elements, and let B be a finite set with n elements. The number of distinct functions mapping A into B is
 - (a) n^m
 - (b) $n!/(n - m)!$
 - (c) $n!$
 - (d) $n!/(m!(n - m)!)$
3. Encapsulating Security Payload (ESP) belongs to which Internet Security Protocol?
 - (a) Secure Socket Layer Protocol
 - (b) Secure IP Protocol
 - (c) Secure Http Protocol
 - (d) Transport Layer Security Protocol
4. Let $f(X) = \text{if } x = 1 \text{ then } 0 \text{ else } [x * f(x - 1) + x^{**2}]$. The value of $f(4)$ is
 - (a) 53
 - (b) 29
 - (c) 50
 - (d) 100
5. The access matrix approach to protection has the difficulty that
 - (a) the matrix, if stored directly, is large and can be clumsy to manage
 - (b) it is not capable of expressing complex protection requirements
 - (c) deciding whether a process has access to a resource is undecidable
 - (d) there is no way to express who has rights to change the access matrix itself

True / False (10 marks)

Answer True or False

6. True or False: The length of the message digest produced by the MD 5 hashing algorithm is 160 bits.
7. True or False: Poor handling of unexpected input is an example of a presentation layer issue related to user interface design.
8. True or False: The IP Network Browser is a tool used for performing SNMP enumeration effectively.
9. True or False: The SYN stealth scan does not complete the TCP handshake, making it less detectable by firewalls.
10. True or False: Windows 7 is known for its robust defenses against buffer-overflow attacks, making it less vulnerable than older systems.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to find the maximum sum of bi-tonic sub-sequence for the given array.
12. Write a function to join the tuples if they have similar initial elements.

End of Paper